Prodigy Anesthesia Quick Review

Fluid/Blood Warmers

What are the methods in which IV fluids are warmed?	IV fluids may be warmed by countercurrent heat exchange, dry heat exchange, fluid immersion, or by placing a part of the fluid circuit close to a separate heater (warming blanket), however this method is less effective. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 230.*
List some of the risks associated with the use of fluid warmers.	Infection, air embolism, pressurized infiltration, heat-induced vessel injury and hemolysis, and the leakage of current into the path of the fluid *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 231.*
The infusion of one liter of crystalloid or one unit of refrigerated blood will decrease mean body temperature to what extent?	For every one unit of refrigerated blood or liter of room temperature crystalloid infused, mean body temperature decreases by 0.25 degrees Celsius. *Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: .* *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 229.*
How is fluid warmed using the dry heat method?	Via a magnetic induction heater, metal blocks or plates, or infrared lamps *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 230.*
What are the disadvantages to the use of simple fluid warmers?	The infusion rate is limited to what can be achieved through gravity and large amounts of air cannot be vented automatically. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 230.*
How do resuscitation fluid warmers differ from simple fluid warmers?	Like simple fluid warmers, resuscitation fluid warmers contain a thermostat and an electric heater, however, resuscitation fluid warmers have the capacity to heat large volumes of fluid infused at high rates. They also have the capacity to stop the infusion of fluid when air is detected through the use of a control system and air detector. *Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 231.*

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What are the disadvantages to the use of water immersion heaters?

The risk for infection is greater with bath solutions, the ability to warm fluids at high infusion rates is limited, there is an increased lag time because of the time required to change the temperature of the bath solution, they are harder to maintain.

Sandberg WS, Urman RD, Ehrenfield JM. The MGH Textbook of Anesthetic Equipment. Philadelphia, PA: Elsevier Saunders; 2011: 230.

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